

KALMAN SZENES

Zurich, Switzerland (US and Hungarian citizen; Swiss C Permit)

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SUMMARY

AI-native computational scientist and software engineer specializing in C++, Python, and high-performance computing. Experienced in GPU kernels, low-latency communication, multi-GPU computation, and numerical software for quantum chemistry. Passionate about agentic software development and autoresearch, performance optimization, and developer tooling.

TECHNICAL SKILLS

Programming	C++, Python, Bash/Shell
Parallel Computing	CUDA, Triton, MPI, OpenMP
Machine Learning	PyTorch, JAX, Distributed Training
Systems & Tooling	Linux, Slurm, Git/GitHub, CMake, Profilers (perf, Nsight Systems/Compute, Intel VTune)
Languages	English (native), French (fluent), Hungarian (native), Russian (native), German (B2)

PROFESSIONAL EXPERIENCE

Graduate Research Assistant — *ETH Zurich, Zurich, Switzerland* **Apr 2023 - Present**

- Implemented spin-free Multireference Coupled Cluster equations in *ORCA* using the *AGE* code generator and demonstrated its scalability on a vitamin B12 model with **809 orbitals**.
- Core developer of *QCMaquis*, a tensor-network package for electronic, vibrational, and vibronic structure and quantum dynamics.
- Contributed to many established quantum chemistry packages, including *ORCA*, *PySCF*, and *OpenMolcas*.

Performance Engineer — *Xanadu Quantum Technologies, Toronto, Canada* **May 2026 - Aug 2026**

- Developed *Backline* in collaboration with AMD, an open-source Python-based MLIR compiler/runtime for low-latency quantum workloads across **CPUs, GPUs, and FPGAs via RDMA**.
- Achieved **2.7 μ s FPGA-to-CPU** and **4.5 μ s FPGA-to-GPU** end-to-end quantum-error-decoding latencies over RoCE v2 with CPU and Triton GPU decoders.
- Co-authored the [technical preprint](#), [company whitepaper](#), and [PennyLane blog post](#) introducing Backline.
- Compiled circuits and performed distributed statevector simulations of **40 qubits on 64 MI355X GPUs** for fluid dynamics.

Performance Engineer — *Algorithmiq, Helsinki, Finland* **Sep 2025 - Jan 2026**

- Achieved over **85% of peak performance** sustained across 8 distributed NVIDIA A100 GPUs by optimizing the hybrid quantum-classical QB-DMRG algorithm for multi-GPU execution.
- Contributed to Algorithmiq's prize-winning [Wellcome Leap Q4Bio](#) submission, earning the team a **USD 2 million prize**.

Research Software Engineer — *CSCS (Swiss National Supercomputing Centre), Zurich, Switzerland* **Mar 2022 - Jul 2022**

- Implemented discontinuous Galerkin schemes for weather and climate simulations using the GT4Py stencil DSL, demonstrating performance portability across CPUs and GPUs.

EDUCATION

PhD Candidate in Quantum Chemistry — *ETH Zurich, Zurich, Switzerland* **Apr 2023 - Expected 2027**

Advisor: Prof. M. Reiher

MSc Computational Science and Engineering — *ETH Zurich, Zurich, Switzerland* **Sep 2020 - Mar 2023**

MSc Thesis: "Tensor Computations on GPUs: From Dense Contractions to Sparse Decompositions"

Advisors: Prof. T. Hoefler, A. Ziogas, T. Ben-Nun

BSc Chemistry and Chemical Engineering — *EPFL, Lausanne, Switzerland* **Sep 2017 - May 2020**

SELECT PERSONAL PROJECTS

MoITUI	Terminal molecular visualizer for trajectories, orbitals, and normal modes using Unicode characters.
MementoChem.com	Convenient web interface for generating input files for common quantum-chemistry calculations.
Blog	Technical blog on lesser-known workflow and development tips.

LIST OF PUBLICATIONS

Unlocking Real-Time Speed for the Entire Quantum Ecosystem with PennyLane and Backline

J. Izaac, B. Lau, L. J. O'Riordan, J. K. L. Lee, M. Malekmohammadi, H.-S. Zheng, C. Doubmbia, L. Neuhaus, T. Germain, T. Ainsworth, M. Zamani Abnili, K. Szenes, S. Shu, J. M. Monsalve Diaz, M. Osama, Y. Eckert, P. Hartke. *Xanadu Whitepaper* (2026).

Python in the front, party in the Backline: compiling quantum workloads across CPUs, GPUs, and FPGAs

J. K. L. Lee, M. Malekmohammadi, H.-S. Zheng, S. Shu, C. Doubmbia, K. Szenes, M. Zamani Abnili, T. Ainsworth, M. Seymour, T. Germain, L. Neuhaus, J. Izaac, L. J. O'Riordan. *arXiv* 2609.09270 (2026).

Boosting Computational Catalysis and Chemical Reactivity with Artificial Intelligence

K. D. Vogiatzis, C. Corminboeuf, A. Nova, K. Jorner, J. Kastner, M. Meuwly, P. Schwaller, V. Bottcher, M. Drosou, E. Fako, H. Hoppe, Z. Ivkovic, N. Iwanjko, D. A. Pantazis, S. P. Schmid, K. Szenes, A. Tetenoire, M. Reiher. *J. Am. Chem. Soc.* 148, 9143-9155 (2026).

Efficient Implementation of the Spin-Free Renormalized Internally-Contracted Multireference Coupled Cluster Theory

K. Szenes, R. Kayal, K. Sivalingam, R. Feldmann, F. Neese, M. Reiher. *J. Phys. Chem. A* 130, 1417-1432 (2026).

QCMaquis 4.0: Multi-Purpose Electronic, Vibrational, and Vibronic Structure and Dynamics Calculations with the Density Matrix Renormalization Group

K. Szenes, N. Glaser, M. Erakovic, V. Barandun, M. Morchen, R. Feldmann, S. Battaglia, A. Baiardi, M. Reiher. *J. Phys. Chem. A* 129, 7549-7574 (2025).

Striking the right balance of encoding electron correlation in the Hamiltonian and the wavefunction ansatz

K. Szenes, M. Morchen, P. Fischill, M. Reiher. *Faraday Discuss.* 254, 359-381 (2024).

Domain-specific implementation of high-order Discontinuous Galerkin methods in spherical geometry

K. Szenes, N. Discacciati, L. Bonaventura, W. Sawyer. *Comput. Phys. Commun.* 295, 108993 (2024).

High-Performance and Programmable Attentional Graph Neural Networks with Global Tensor Formulations

M. Besta, P. Renc, R. Gerstenberger, P. Sylos Labini, A. Ziogas, T. Chen, L. Gianinazzi, F. Scheidl, K. Szenes, A. Carigiet, P. Iff, G. Kwasniewski, R. Kanakagiri, C. Ge, S. Jaeger, J. Was, F. Vella, T. Hoefler. *Proceedings of Supercomputing '23* 1-16 (2023).

SUMMER/WINTER SCHOOLS

Winter School on Theoretical Chemistry: Electronic Structure Theory — Helsinki, Finland

Dec 2025

🏆 **Best Poster Prize:** "Efficient Implementation of the Spin-Free Renormalized Internally-Contracted Multireference Coupled Cluster Theory".

Modern Wavefunction Methods — Pisa, Italy

Aug 2023

Poster: "Tensor Computations on GPUs: From Dense Contractions to Sparse Decompositions."

International Summer School on High Performance Computing — PRACE, XSEDE, RIKEN, SciNet, EPCC and Pawsey, Atlanta, USA

Jun 2023

Poster: "Tensor Computations on GPUs: From Dense Contractions to Sparse Decompositions."

Summer School in Effective HPC and Data Analytics with GPUs — CSCS, Virtual, Switzerland

Aug 2021

SELECT CONFERENCES AND WORKSHOPS

First Orca Developer Workshop — Online

Mar 2026

Talk: "Developing Spinfree Many-Body Residual Equations in Orca."

Workshop on Tensor Contraction Library Standardization — CECAM, Toulouse, France

Sep 2025

Talk: "The Density Matrix Renormalization Group in Quantum Chemistry"

World Association of Theoretical and Computational Chemists — Oslo, Norway

Jun 2025

Poster: "Multi-State CASPT2 with a DMRG Reference Wavefunction."

OpenMolcas Developers' Workshop — Munich, Germany

Jun 2025

Talk: "QCMaquis 4.0: Multi-Purpose Electronic, Vibrational, and Vibronic Structure and Dynamics Calculations with the Density Matrix Renormalization Group."

Workshop on Tensor Contraction Library Standardization — CECAM, Toulouse, France

Sep 2024

Poster: "Implementation Considerations for the Second-Generation Density Matrix Renormalization Group."

Faraday Discussion on Correlated Electronic Structure — Royal Society of Chemistry, London, UK

Jun 2024

Paper: "Striking the right balance of encoding electron correlation in the Hamiltonian and the wavefunction ansatz."

TEACHING EXPERIENCE

Teaching assistant at ETH Zurich for: *Advanced Quantum Chemistry, Advanced Statistical Physics, Statistical Physics and Computer Simulation, Informatics, Quantum Chemistry Practical Course, Programming Techniques for Scientific Simulations*